

RESEARCH ARTICLE

Sediment-Resistant Marine Antifouling Coating Based on Amphiphilic Copolymer Brushes Bearing Sulfobetaine and Isobornyl Pendants

Jihwan Do, Jinwoo Lee, Jong Hwan Suh,* and Sung Min Kang*

For marine antifouling coatings to be practically useful, resistance to both marine fouling organisms and inorganic marine sediments is essential. Amphiphilic copolymer coatings, which integrate hydrophilic and hydrophobic components, are known to impart such dual antifouling properties to solid substrates. In these systems, the hydrophilic component functions as a physical barrier to inhibit biological fouling, while the hydrophobic component prevents the adsorption of marine sediments. Based on this principle, various amphiphilic copolymers have been developed for marine antifouling applications. To effectively minimize both biological adhesion and sediment accumulation, in addition to selecting appropriate hydrophilic and hydrophobic monomers, optimizing their relative composition is crucial. In this study, amphiphilic copolymer coatings composed of sulfobetaine methacrylate (SBMA) and isobornyl methacrylate (IBMA) are synthesized via surface-initiated atom transfer radical polymerization. Through a systematic investigation, the optimal SBMA-to-IBMA ratio is determined to achieve a balanced antifouling performance. Notably, coatings prepared with an SBMA:IBMA feed molar ratio of 1:1 effectively suppress both marine diatom adhesion and sediment adsorption. Surface free energy analysis reveals that superior antifouling performance correlates with a lower surface free energy. These findings underscore the design parameters that should be considered in the development of high-performance amphiphilic copolymer coatings for marine antifouling applications.

1. Introduction

As environmentally friendly materials for marine antifouling applications, amphiphilic copolymers have recently attracted significant attention owing to their dual hydrophobic and hydrophilic nature that provides a synergistic approach to combat biofouling.^[1–20] This interest stems from the shortcomings observed for previously developed antifouling polymers. For instance, hydrophilic polymers, such as zwitterionic polymers and poly(ethylene glycol) (PEG), have been widely synthesized and applied in marine environments to provide excellent antifouling performance against a broad range of fouling organisms.^[21–30] Jiang et al. reported the preparation of zwitterionic sulfobetaine polymer coatings on solid substrates, which exhibited ultralow fouling behavior against diatoms and *Ulva* spores.^[23] Similarly, the strong antifouling properties of PEG-coated surfaces have been extensively reported.^[29,31] The antifouling efficacy of these hydrophilic polymers is primarily attributed to the formation of hydration layers on the coated surfaces.^[32]

These hydration layers, formed along zwitterionic polymer chains or PEG segments, act as physical barriers that effectively inhibit the adhesion of marine fouling organisms.^[30] However, recent studies have revealed the drawbacks of hydrophilic polymers for marine antifouling applications. Marine environments contain not only biological fouling organisms but also inorganic sediments. Although hydrophilic surfaces are effective at resisting biological adhesion, they are prone to sediment adsorption, which can contaminate polymer-coated surfaces and diminish antifouling performance.^[33,34] In contrast, hydrophobic polymer coatings are generally less susceptible to sediment adsorption; however, these coatings tend to promote the adhesion of marine organisms. These opposing limitations have prompted the development of alternative strategies—most notably, the synthesis of amphiphilic copolymer coatings—for integrating the benefits of both hydrophilic and hydrophobic components to achieve balanced and durable marine antifouling performance.

J. Do, J. Lee, S. M. Kang
Department of Chemistry
Chungbuk National University
Cheongju, Chungbuk 28644, Republic of Korea
E-mail: smk16@chungbuk.ac.kr

J. H. Suh
Department of Management Information Systems & BERI
Gyeongsang National University
Jinju, Gyeongnam 52828, Republic of Korea
E-mail: jonghwan.suh@gnu.ac.kr

 The ORCID identification number(s) for the author(s) of this article can be found under <https://doi.org/10.1002/admi.202500491>

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DOI: 10.1002/admi.202500491

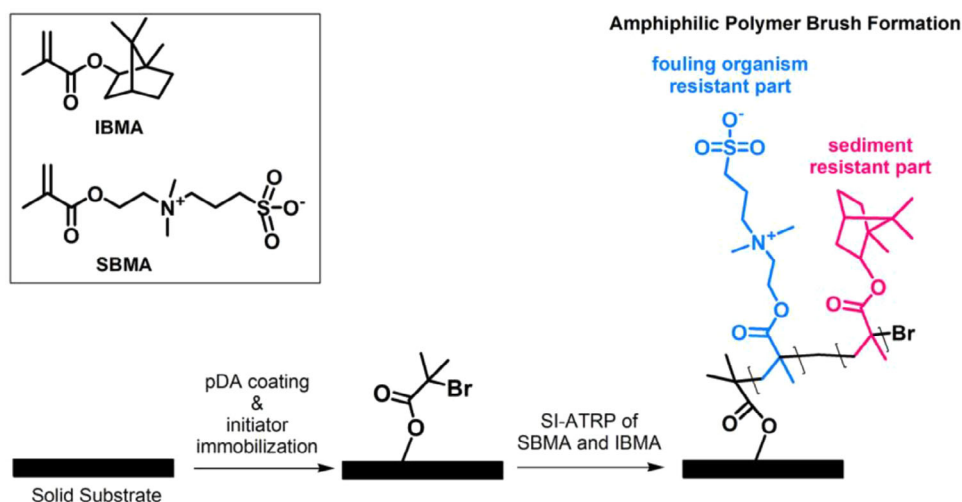


Figure 1. Schematic of amphiphilic polySBIB brush formation on solid substrates.

As the hydrophobic component of marine antifouling amphiphilic copolymers, fluorine-containing monomers have been widely used because they offer excellent hydrophobicity and effectively reduce surface free energy.^[4,11–15,17,35] However, their use is limited owing to the increasing environmental concerns, thus driving interest in more environmentally benign hydrophobic monomers for the synthesis of amphiphilic copolymers. Notable alternatives include butyl methacrylate, isobornyl methacrylate (IBMA), and dicyclopentenyl ether acrylate.^[5,6,15,36] IBMA is of particular interest due to its borneol core structure, which is derived from renewable resources such as lavender and chamomile.^[15] Along with its sustainable origin, borneol is known for its antibacterial properties, thereby making IBMA a superior choice over other synthetic hydrophobic monomers for amphiphilic copolymer coatings.^[15] This was also demonstrated by Liu et al., who synthesized amphiphilic copolymers consisting of IBMA, sulfobetaine methacrylate (SBMA), and trifluoroethyl methacrylate (TFEMA), which showed excellent antifouling performance against proteins, bacteria, and diatoms.^[15]

Despite this progress, a systematic investigation into the relationship between the ratio of hydrophilic and hydrophobic components, specifically SBMA and IBMA, and the resulting marine antifouling performance is still lacking. Furthermore, most antifouling evaluations have exclusively focused on biological adhesion, making it difficult to assess the performance in environments containing marine sediments. Therefore, in this study, we aim to systematically synthesize amphiphilic copolymer coatings with varying ratios of isobornyl and sulfobetaine groups and evaluate their antifouling performance against both marine sediment and marine diatoms. Through this approach, we also aim to propose an optimized amphiphilic copolymer coating with well-balanced marine antifouling properties.

2. Results and Discussion

This study focused on the preparation of copolymer coatings on solid substrates with controlled amphiphilicity. Among various methods for polymer coating, surface-initiated, atom transfer radical polymerization (SI-ATRP)—well-known for its com-

patibility with a wide range of monomers—was employed to prepare all copolymer brushes via a standardized approach.^[26] Consequently, both copolymers and homopolymers of SBMA and IBMA could be synthesized on solid substrates via the same approach. The initiator coating involved the following two-step process: a polydopamine (pDA) pre-coating followed by Zr(IV)-mediated immobilization of a phosphonate-terminated initiator onto the pDA-coated surface.^[13] Owing to the material-independent coating properties of pDA and its strong affinity for forming coordination bonds with metal ions, the initiator was effectively immobilized on the substrate.^[37,38] Specifically, the phosphonate-terminated initiator was immobilized via Zr(IV)-mediated multiple coordination bonds between the initiator and pDA layer (**Figure 1**). Amphiphilic copolymer brushes bearing sulfobetaine and isobornyl side chains were subsequently grown on the initiator-coated surface using SI-ATRP. To systematically investigate the effect of IBMA content on the amphiphilicity and antifouling properties of the copolymer brushes, we synthesized copolymer brushes with various SBMA:IBMA feed molar ratios, namely 8:2, 7:3, 5:5, 3:7, and 2:8. These copolymer brushes were denoted as polySBIB(X), where X represents the mole fraction of SBMA. For example, polySBIB(0.8) corresponds to a copolymer brush synthesized with an SBMA:IBMA feed molar ratio of 8:2.

The formation of various polymer brushes on solid substrates was confirmed via X-ray photoelectron spectroscopy (XPS). As shown in **Figure 2**, all the SBMA-containing polymer brushes, polySBMA and polySBIB, exhibited both S 2p and N 1s peaks corresponding to the sulfonate and ammonium groups of SBMA. In contrast, these peaks were absent in the polyIBMA brush, as expected, because of the lack of SBMA units. The SBMA content in polySBIB brushes was further evaluated by analyzing the surface atomic compositions. A decreasing trend in the S 2p peak intensity was observed as the SBMA mole fraction decreased from 0.8 to 0.2—3.77% for polySBIB(0.8), 3.26% for polySBIB(0.7), 3.36% for polySBIB(0.5), 1.42% for polySBIB(0.3), and 2.21% for polySBIB(0.2). These results confirm the successful incorporation of SBMA into the copolymer brushes and reflect the expected variation in the SBMA content based on feed composition. Previously, Liu et al. demonstrated that copolymer coatings composed

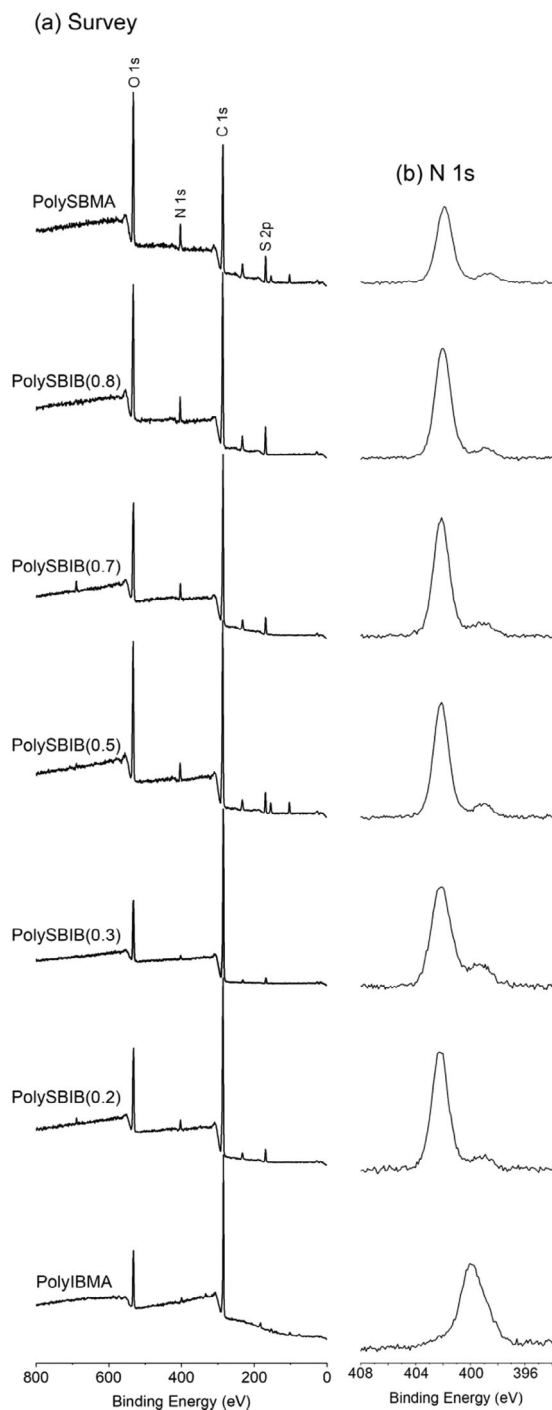


Figure 2. a) Survey and b) high-resolution (N 1s) X-ray photoelectron spectra of Si/SiO₂ substrates coated with polySBMA, polySBIB, and polyIBMA.

of IBMA, SBMA, and TFEMA exhibit superior marine antifouling performance compared to sulfobetaine homopolymers.^[15] They performed molecular dynamics (MD) simulations, which revealed that hydrophobic units promote the migration of hydrophilic SBMA units toward the outer surface. Although Liu et al. conducted their antifouling tests under sediment-free con-

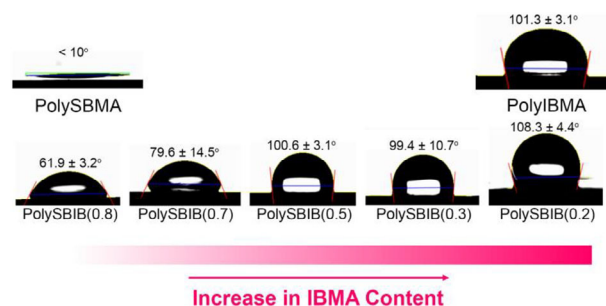


Figure 3. Water contact angles of Si/SiO₂ substrates coated with polySBMA, polySBIB, and polyIBMA.

ditions, this phenomenon suggests that the extent of SBMA exposure at the surface of certain polySBIBs in this study may be greater than expected. For instance, XPS analysis revealed that the sulfur (S 2p) content of polySBIB(0.2) was higher than that of polySBIB(0.3). This result can be explained by the enhanced migration of SBMA units to the outer surface, promoted by the sufficient presence of IBMA in polySBIB(0.2).

Identifying the optimal level of amphiphilicity for marine antifouling applications was a primary objective of this study. To this end, we evaluated the wettability of the prepared polymer brushes. As the mole fraction of SBMA in the copolymer brushes was varied from 0.8 to 0.2, surfaces with a broad range of wettability were produced, with the water contact angle (WCA) ranging from the minimum value of 62° to the maximum of 108° (Figure 3). Notably, when the mole fraction of IBMA reached 0.5, the WCA approached 100°, and no significant increase in the WCA was observed beyond this point.

The surface morphologies of the polymer brush-coated surfaces were characterized using atomic force microscopy (AFM). As shown in Figure 4, the polySBMA, polySBIB(0.8), polySBIB(0.7), and polyIBMA brushes exhibited relatively smooth surfaces, with root-mean-square (RMS) roughness values of 2.7, 2.1, 2.1, and 3.3 nm, respectively. Although the increase was not substantial, a remarkable increase in the RMS roughness was observed in the copolymer brushes with SBMA mole fractions below 0.5, with RMS roughness values of 7.2 nm for polySBIB(0.2), 13.7 nm for polySBIB(0.3), and 12.7 nm for polySBIB(0.5). This effect is attributable to microphase separation between the hydrophilic SBMA and hydrophobic IBMA components in the copolymer brushes with relatively low coating thicknesses (Table S1), as all the polymer layers were thinner than 85 nm. These results indicate that incorporating IBMA into thin polySBMA brushes does not markedly influence the surface roughness in our system.

The marine antifouling properties of the polymer brushes were evaluated using marine diatoms in the presence of marine sediments. Among the copolymer brushes, polySBIB(0.5) and polySBIB(0.3) effectively suppressed diatom adhesion, whereas the others showed limited antifouling performance (Figure 5). Quantitatively, diatom adhesion decreased to 62 and 130 cells on polySBIB(0.5) and polySBIB(0.3), corresponding to 77% and 53% reductions, respectively, relative to polySBMA (275 cells). In contrast, severe diatom attachment was observed on both polySBMA and polyIBMA brushes. Given the contrasting surface wettabilities of these brushes, this finding suggests that diatom adhesion

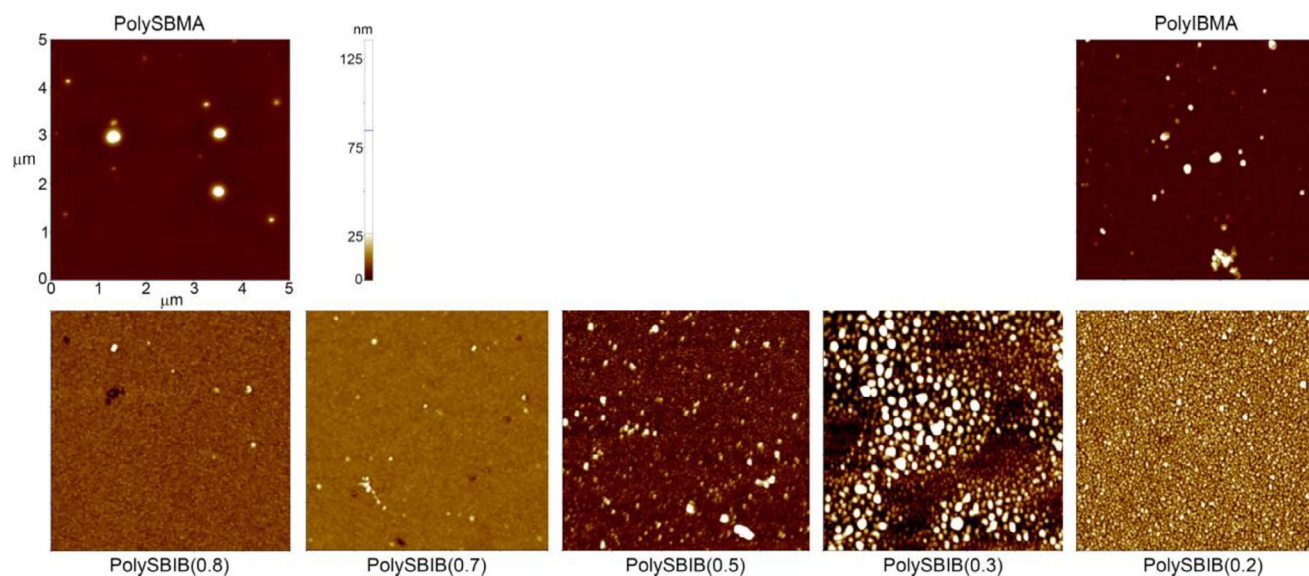


Figure 4. AFM images of Si/SiO₂ substrates coated with polySBMA, polySBIB, and polyIBMA.

was governed by different mechanisms across the different surfaces. For polySBMA, severe diatom adhesion is attributed to sediment adsorption, whereas for polyIBMA, it is likely due to the diatoms' preference for hydrophobic surfaces. This explanation is supported by the results of the antifouling tests conducted in the absence of marine sediments (Figure S1). Under these conditions, diatom adhesion behavior correlated strongly with coating hydrophobicity; the attached diatom density increased with the incorporation of more hydrophobic monomers (i.e., IBMA) into the amphiphilic copolymer brushes. Consistent with this trend, severe diatom adhesion was observed on the polyIBMA sample.

The following two additional experiments were conducted to elucidate the underlying reason why polySBIB(0.5) exhibited the best marine antifouling performance: (1) surface free energy calculation, and (2) an adsorption test using marine sediment alone. The surface free energy (γ_s) was calculated using the Owens–Wendt geometric mean equation,^[39] which divides the total surface free energy into dispersive (γ_s^D) and polar (γ_s^P) components as follows:

$$(1 + \cos \theta) \gamma_L = 2\sqrt{\gamma_s^D \gamma_L^D} + \sqrt{\gamma_s^P \gamma_L^P} \quad (1)$$

where θ is the measured contact angle of the liquid on the surface, and γ_L is the surface tension of the liquid. The total surface free energy (γ_s ; $\gamma_s = \gamma_s^D + \gamma_s^P$) for each surface was determined by measuring the contact angles of two probe liquids, namely water and diiodomethane (CH₂I₂). The polySBMA-coated surface exhibited a surface free energy of 74.6 mJ m⁻², whereas the other polymer brush-coated surfaces bearing isobornyl pendants showed surface free energies in the range of 40–50 mJ m⁻² (Table 1). The surface free energy significantly decreased as the mole fraction of SBMA in polySBIB was reduced from 0.8 to 0.5. However, when the mole fraction of SBMA was further reduced to 0.2, no significant change in the surface free energy was observed. Additionally, the surface free energy of polySBIB(0.5) was

not significantly different from that of polyIBMA. This indicates that incorporating IBMA into polySBMA up to a mole fraction of 0.5 is sufficient to effectively reduce the surface free energy (Figure 6).

On the basis of this observation, a marine sediment adsorption experiment was conducted. As shown in Figure S2, polymer brushes with higher SBMA content (e.g., polySBIB(0.8)) exhibited substantial adsorption of large, dark-colored sediments, likely because of the strong interaction between SBMA and marine sediments. At SBMA mole fractions of 0.7 and 0.5, sediment adsorption was remarkably reduced and remained at a similar level. Below this range, further reductions were observed for polySBIB(0.3), polySBIB(0.2), and polyIBMA. However, despite the apparent decrease in sediment adsorption, no significant decrease in the surface free energy was observed compared to that of polySBIB(0.5). This discrepancy may be attributed to the presence of submicron-sized sediments that cannot be reliably detected using optical microscopy. Based on the surface free energy results, polySBIB(0.3), polySBIB(0.2), and polyIBMA are considered to induce sediment adsorption at levels comparable to those of polySBIB(0.5). Collectively, these findings suggest that the reduction in surface free energy caused by IBMA effectively inhibits marine sediment adsorption, particularly when the SBMA mole fraction falls below 0.5. However, a lower SBMA content also leads to increased diatom adhesion. Thus, a balanced composition of SBMA and IBMA—specifically, polySBIB(0.5)—is apparently optimal for simultaneously minimizing the attachment of both marine sediments and marine diatoms.

The advantage of our approach to constructing polySBIB copolymer brushes on solid substrates lies in its broad applicability, enabled by the use of pDA precoating.^[38] In this context, a polySBIB copolymer brush was prepared on stainless steel (SS) substrates via the same method. SS was selected as an additional model substrate owing to its widespread use in marine vessel hulls.^[29] The uncoated SS surface was first precoated with pDA, followed by immobilization of the initiator through

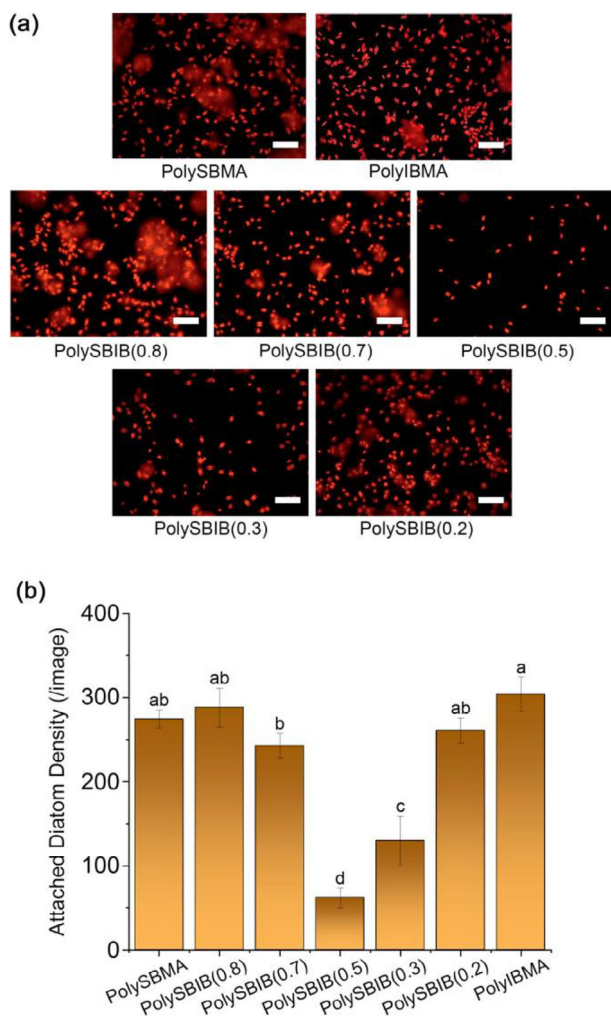


Figure 5. a) Fluorescence images and b) quantification of diatoms attached to different coatings in the presence of marine sediment (image area = 0.115 mm²). Scale bars represent 50 μ m.

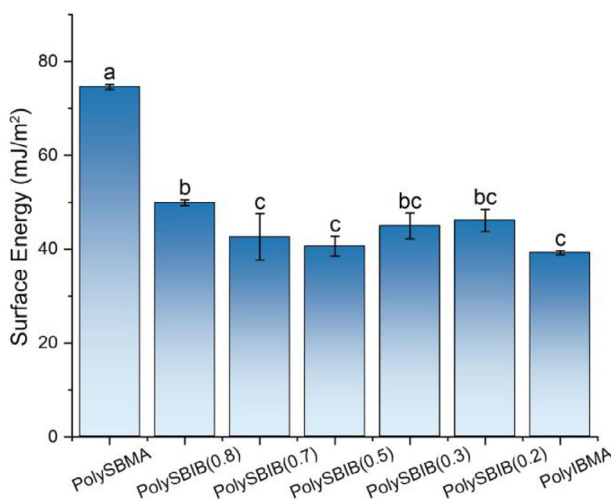


Figure 6. Surface free energies of the different polymer brushes.

Table 1. Static CH₂I₂ contact angles and surface free energies of different polymer brushes.

Polymer brush	Diodomethane contact angle [°]	Surface free energy [mJ m ⁻²]
PolySBMA	33.9 ± 3.7	74.6 ± 0.5
PolySBIB(0.8)	37.4 ± 3.3	49.9 ± 0.6
PolySBIB(0.7)	41.2 ± 2.6	42.6 ± 4.9
PolySBIB(0.5)	38.3 ± 4.5	40.7 ± 2.1
PolySBIB(0.3)	31.0 ± 2.6	45.0 ± 2.7
PolySBIB(0.2)	30.5 ± 2.8	46.2 ± 2.4
PolyIBMA	41.2 ± 0.5	39.3 ± 0.5

Zr(IV)-mediated crosslinking between the pDA layer and phosphonate-terminated initiator. Among the various polySBIB compositions, polySBIB(0.5), which exhibited the best marine antifouling performance on Si/SiO₂ substrates, was selected for coating the SS surface. Following the formation of the polySBIB(0.5) brush, the modified SS substrates were characterized using WCA goniometry and XPS. As shown in Figure 7, the WCA increased from 64.1° ± 0.6° to 96.5° ± 8.6° after the formation of the polySBIB(0.5) brush, clearly indicating the successful growth of the amphiphilic copolymer brush on the SS substrate (Figure 7a). In addition, XPS analysis revealed the appearance of new characteristic peaks corresponding to sulfur (S 2s and S 2p), attributed to the sulfonate groups in SBMA, and a concurrent decrease in the substrate peaks for Fe 2p. These findings further confirmed the successful formation of polySBIB(0.5) brush on the SS surface (Figure 7b).

Prior to the marine diatom adhesion test, sediment resistance was evaluated (Figure S3). The resistance of the polySBIB(0.5) brushes against sediment adsorption was also confirmed on the SS substrates. In contrast to uncoated SS, which exhibited severe sediment adsorption, the polySBIB(0.5) brushes remarkably suppressed sediment accumulation. The marine antifouling performance of the polySBIB(0.5) brushes on the SS surfaces was then evaluated. In addition to the adhesion assays previously conducted on Si/SiO₂ substrates, a stability test was performed using the polySBIB(0.5) brushes on SS as marine vessels are typically submerged in seawater for extended periods. To assess the stability of the copolymer brush under marine-like conditions, we incubated polySBIB(0.5)-coated SS substrates in artificial seawater for 1 and 2 weeks. After incubation, the samples were subjected to a standard marine diatom adhesion test in the presence of marine sediments. Uncoated SS served as the control to compare the fouling behavior of marine diatoms. The uncoated SS surface exhibited severe diatom attachment (Figure 8). In contrast, the as-prepared polySBIB(0.5) brush (i.e., without immersion in artificial seawater) showed an 80% reduction in diatom adhesion compared to uncoated SS. However, immersion in artificial seawater for 1 and 2 weeks gradually diminished the antifouling performance of the brush. The attached diatom density increased from 42 (no immersion) to 76 (1 week) and 111 (2 weeks). Despite this decline in performance, the polySBIB(0.5) brush achieved ≈50% reduction in diatom adhesion compared to uncoated SS even in the presence of marine sediments. The decrease in the antifouling performance is attributable to the hydrolytic

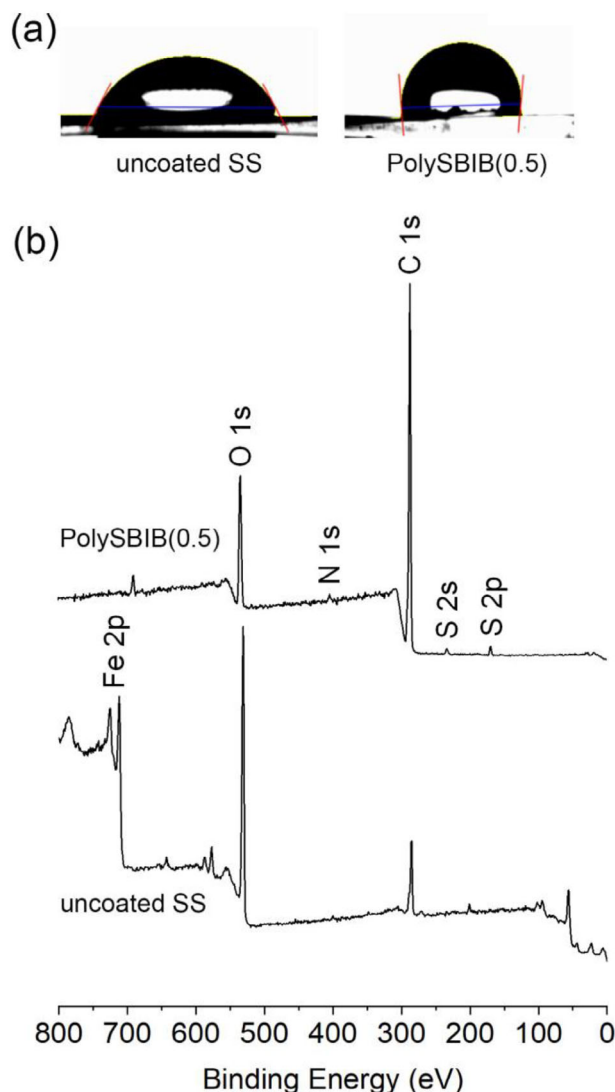


Figure 7. a) WCA images and b) X-ray photoelectron spectra of uncoated and polySBIB(0.5)-coated stainless steel (SS) substrates.

degradation of the ester bonds in the repeating units of the copolymer. Ester bonds are susceptible to hydrolysis, particularly under high-salinity environments such as seawater, and their cleavage may compromise the structural integrity and functionality of the copolymer brushes over time. To validate this explanation, we reanalyzed the surface characteristics of the polySBIB(0.5)-coated SS substrates after 2 weeks of immersion in artificial seawater using WCA measurement and XPS (Figure S4 and Table S2). The WCA decreased to $76.8^\circ \pm 3.4^\circ$, accompanied by changes in the surface atomic composition (Figure S4a). Specifically, a decrease in the C 1s peak intensity (82.2% $\rightarrow$ 80.1%) and an increase in the O 1s peak intensity (15.41% $\rightarrow$ 17.19%) were observed, whereas the peak intensities of N 1s and S 2p remained largely unchanged (Figure S4b). These results suggest that a portion of the carbon-rich hydrophobic isobornyl pendant units was cleaved from the polySBIB(0.5) main chain during immersion, thereby reducing the carbon content and increasing the hydrophilicity.

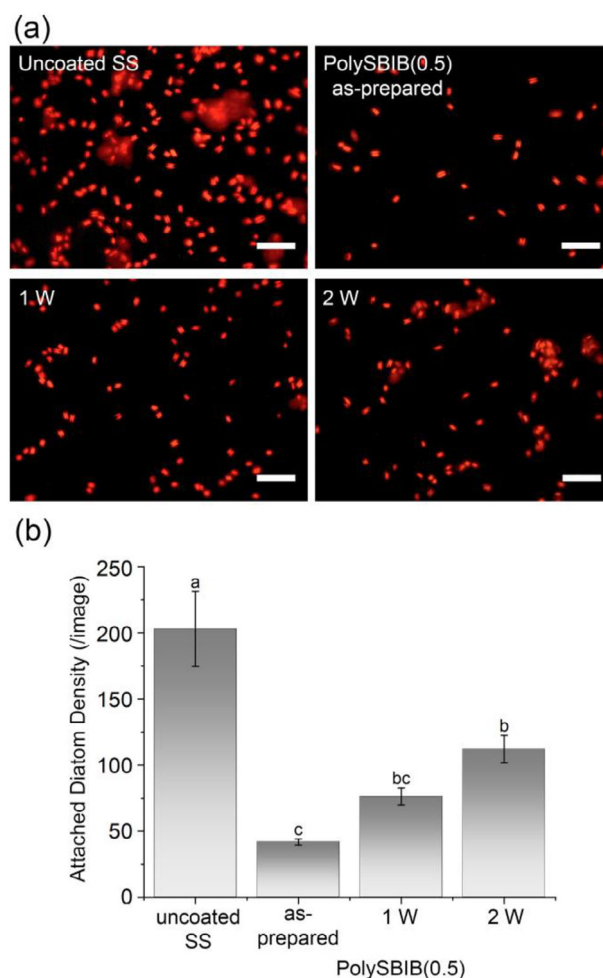


Figure 8. a) Fluorescence images and b) quantification of diatoms attached to SS substrates coated with the polySBIB(0.5) brush after immersion in artificial seawater for 1 and 2 weeks (image area = 0.115 mm²). The scale bars represent 50 μ m.

3. Conclusion

In this study, we developed amphiphilic copolymer coatings with dual marine antifouling properties, comprising hydrophilic sulfobetaine and hydrophobic isobornyl pendant groups. The ratio between these two components was systematically varied to determine the optimal composition for balanced marine antifouling performance. Among the formulations studied, the copolymer brush with a 1:1 feed molar ratio of sulfobetaine and isobornyl pendants exhibited the best surface properties, effectively resisting both marine diatom adhesion and sediment adsorption. To elucidate the origin of this enhanced antifouling performance, we analyzed the surface free energy of the copolymer brushes and observed a correlation between the lower surface free energy and improved antifouling efficacy. Our findings offer important design criteria for the development of the next generation of environmentally friendly marine antifouling materials.

4. Experimental Section

Materials: Tris(hydroxymethyl)aminomethane (Trizma base, 99%), tris(hydroxymethyl)aminomethane hydrochloride (Trizma HCl, 99%), dopamine hydrochloride (DA, 98%), zirconyl chloride octahydrate ($\text{ZrOCl}_2 \cdot 8\text{H}_2\text{O}$, 98%), [2-(methacryloyloxy)ethyl]dimethyl-(3-sulfopropyl)ammonium hydroxide (SBMA, 95%), N,N,N',N'',N''-pentamethyldiethylenetriamine (PMDETA, 99%), phosphate-buffered saline (PBS), and IBMA were purchased from Sigma-Aldrich. 2,2,2-trifluoroethanol (TFE, >99%), zirconium(IV) chloride (ZrCl_4), copper(I) bromide (CuBr , 98%), and copper(II) bromide (CuBr_2 , 99%) were obtained from Alfa Aesar. [11-[(2-Bromo-2-methylpropanoyl)oxy]undecyl]phosphonic acid (initiator, >98%), L-ascorbic acid (AA), and 2,2'-bipyridyl (bpy) were purchased from TCI. The marine benthic diatom *Amphora coffeaeformis* was acquired from the Korea Marine Microalgae Culture Center.

Initiator Coating on Solid Substrates: The solid substrates (Si/SiO_2 and SS) were initially cleaned by sonication in ethanol for 5 min, followed by drying under a nitrogen gas stream. Subsequently, the substrates were immersed in a 10 mM DA solution prepared in Tris buffer (pH 8.4) for 4 h and then rinsed thoroughly with deionized (DI) water. The resulting pDA-coated substrates were treated with a 5 mM ethanol solution of ZrOCl_2 for 10 min, rinsed sequentially with ethanol and DI water, and dried under nitrogen gas. Finally, the substrates were incubated in a 1 mM ethanol solution of the initiator for 10 min, rinsed again with ethanol and DI water, and dried under a nitrogen gas stream.

Formation of Poly(SBMA-co-IBMA) and Poly(IBMA) Brushes on Si/SiO_2 Substrates via Surface-Initiated Atom Transfer Radical Polymerization: Before use, IBMA was purified by passing it through a basic alumina syringe column to remove the polymerization inhibitors. The monomers, SBMA and IBMA, were dissolved in varying ratios in a mixture of TFE and methanol (TFE/MeOH = 2:1 v/v). A 1.5 mL aliquot of the monomer solution was then transferred to a glass vial containing the initiator-coated solid substrate (Table S3). The solution was degassed by purging with nitrogen for 25 min. Separately, a copper catalyst solution (1.5 mL) consisting of CuBr (14.98 mg), CuBr_2 (4.62 mg), and PMDETA (27.86 mg) was prepared and degassed with nitrogen for 25 min. The degassed copper/ligand solution was then added to the monomer mixture to initiate the polymerization; the reaction was performed at 70 °C for 16 h. After completion, the substrates were thoroughly washed with TFE and MeOH and dried under a nitrogen gas stream.

Formation of Poly(SBMA) Brushes on Si/SiO_2 Substrates via Surface-Initiated Activators Regenerated by Electron Transfer Atom Transfer Radical Polymerization: Two aqueous stock solutions of bpy (1.16 mg mL^{-1}) and CuBr_2 (0.276 mg mL^{-1}) were prepared and combined in a 1:1 volume ratio. A 0.33 mL aliquot of the resulting complex solution was added to 2.87 mL of SBMA solution (0.58 g mL^{-1}). The initiator-coated solid substrates were then immersed in the reaction mixture. Subsequently, 0.13 mL of an AA solution (13.26 mg mL^{-1}) was introduced into the reaction to initiate polymerization. After 1 h, the substrates were thoroughly rinsed with DI water and PBS and dried under a nitrogen gas stream.

Dynamic Marine Diatom Adhesion Test with Sediment: A suspension of marine diatoms was prepared at a chlorophyll concentration of 3 $\mu\text{g mL}^{-1}$. Natural sand sediment collected from the test site at Port Boryeong, South Korea (36°20'15.1"N 126°31'57.5"E) was added to the suspension at a concentration of 37.5 g L^{-1} . A 13 mL aliquot of the resulting mixture was applied to the solid substrates and incubated using a MaXshakeRP2 reciprocating shaker (Daihan Scientific Co.) at 20 °C and 50 rpm for 24 h. Following incubation, the solid substrates were gently rinsed with artificial seawater. The diatom fouling behavior on the substrate surfaces was analyzed using fluorescence microscopy.

Diatom Detection and Counting using YOLOv9: To quantify the diatoms in the fluorescence images obtained in this study, a diatom detection model was developed based on the You Only Look Once, version 9 (YOLOv9) algorithm. This model was designed for single-class detection of red-fluorescent diatoms. The YOLO architecture was selected owing to the wide recognition of its superior performance in object detection

tasks.^[40,41] Details regarding the model development and its application for diatom counting are provided in the Supporting Information.

Characterization of Polymer-Coated Substrates: XPS was conducted using a NEXSA G2 spectrometer (Thermo Scientific, Waltham, USA) equipped with an Al K α X-ray source. The coating thicknesses on the solid substrates were measured using an alpha-SE spectroscopic ellipsometer (J.A. Woollam Co., Lincoln, USA) over a wavelength range of 390–900 nm, at an incidence angle of 70°. Data analysis was performed using the Cauchy model in the CompleteEASE software, with the refractive index defined as $n(\lambda) = 1.45 + 0.01/\lambda^2$. Static WCA measurements were carried out using a Phoenix-10 system (Surface Electro Optics, Suwon, South Korea) to assess the hydrophobicity of the coatings. AFM imaging was performed in the non-contact mode using an XE-7 microscope (Park Systems, Suwon, South Korea). Fluorescence imaging was performed using an optical microscope equipped with a fluorescence module (LV100ND; Nikon, Tokyo, Japan). The organic molecules used in this study were analyzed by liquid chromatography–mass spectrometry (Agilent G6135BA, NFEC-2024-11-300749).

Statistical Analysis: All experiments were conducted in triplicate using independently prepared samples. Data are presented as the mean $\pm$ standard deviation. Statistical analysis was performed using one-way analysis of variance, followed by Tukey's post-hoc test for multiple comparisons, with a significance level of $\alpha = 0.05$.

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

Acknowledgements

J.D. and J.L. contributed equally to this work. This research was supported by the Basic Science Research Program through the National Research Foundation of Korea (NRF), funded by the Ministry of Science and ICT (RS-2022-NR069220 to S.M.K. and NRF-2021R1F1A1063681 to J.H.S.).

Conflict of Interest

The authors declare no conflict of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Keywords

amphiphilic copolymer, diatom adhesion, isobornyl, sediment adsorption, sulfobetaine

Received: May 30, 2025
Revised: September 18, 2025
Published online: October 8, 2025

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